Comparative Analysis of Machine Learning and Deep Learning Techniques for Handwritten Digit Recognition

Final Data Science Project

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Problem and project objectives

The problem

Different people write the same digit in different ways.

Learn a label from 0 to 9 from numerical image pixels.

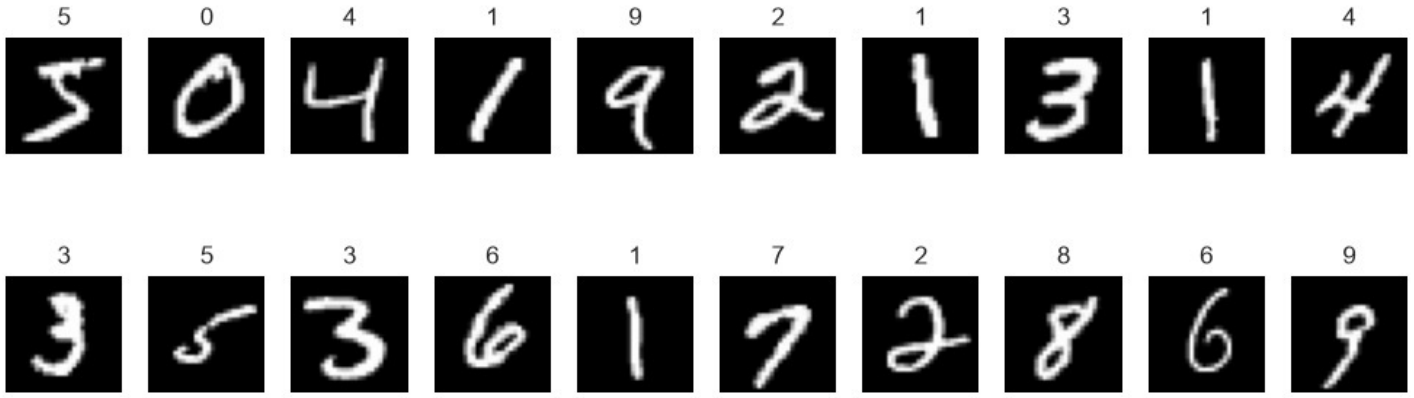
Project objectives

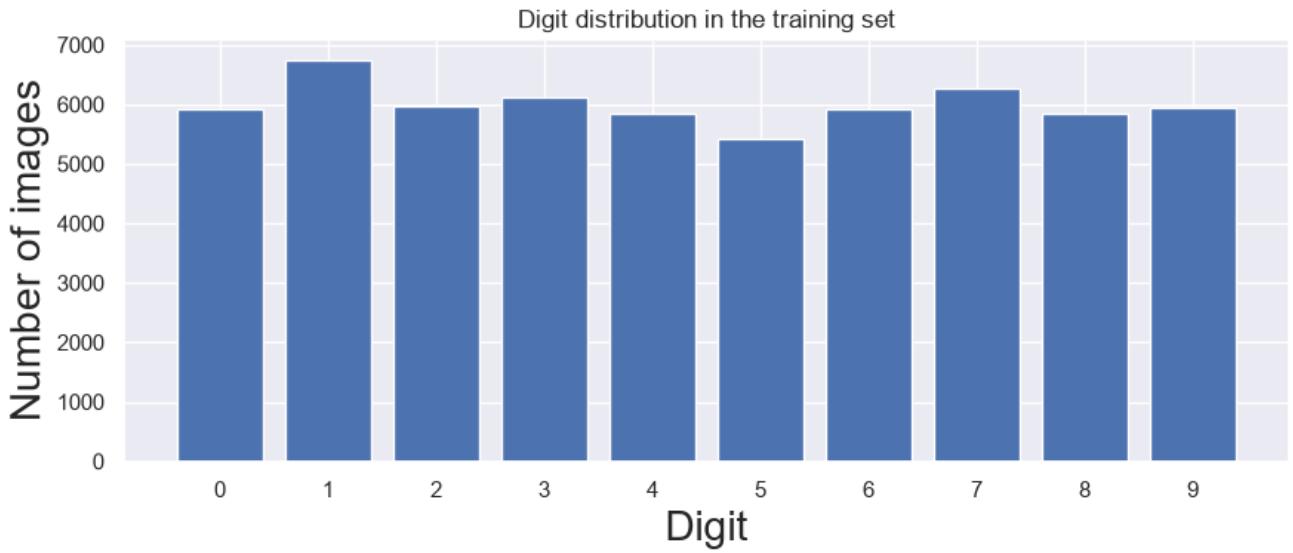
Explore and prepare MNIST.

Compare Logistic Regression with a simple dense network.

Assess frozen models using separate test data.

MNIST and exploratory analysis





Original training data only. Top: first twenty images. Bottom: class counts. Source: saved Stage 1 figures

70,000 images

60,000 official training
10,000 official test

28 × 28 grayscale pixels
Integer labels from 0 to 9

Training classes
1: 6,742 images
5: 5,421 images

Approximately balanced

Data preparation and separation

Partition	Images	Purpose
Training	54,000	Fit model parameters
Validation	6,000	Select network width
Official test	10,000	Final frozen assessment

Seed 42. Shuffle training indices once. Save and reuse the same split.

float32(pixel) / 255 Pixels: 0–255 to 0–1
Logistic Regression: 784 features Network: 28 × 28, then Flatten

Classroom patterns: Untitled20.ipynb, cells 6–7. Untitled19.ipynb, cell 3. Indices remain paired with labels

Logistic Regression

784 normalised pixel features

Class score: z_k = sum_j(w_kj x_j) + b_k
Multinomial probabilities select the most likely digit.

Scikit-learn LogisticRegression
C = 1.0 solver = 'lbfgs' max_iter = 1000

235 solver iterations. No convergence warnings.

Bank_Logistic_Regression.ipynb, cells 5–6. Required adaptation: Scikit-learn multiclass API replaces binary Statsmodels

Classroom-style dense neural networks

Input and Flatten	28 × 28 image becomes 784 values
Hidden layers	Dense 50 + Dense 50, or Dense 100 + Dense 100 ReLU: max(0, z)
Output	Exactly 10 Softmax probabilities for digits 0–9
Training	Adam and sparse categorical cross-entropy 5 epochs, batch size 100, initialization seed 42

Untitled19.ipynb, cells 7–11. Corrected lesson output width to 10. Weights and biases: Untitled18.ipynb, cells 5–7

Validation experiment and model selection

Same split, seed, optimiser, loss, epochs and batch size

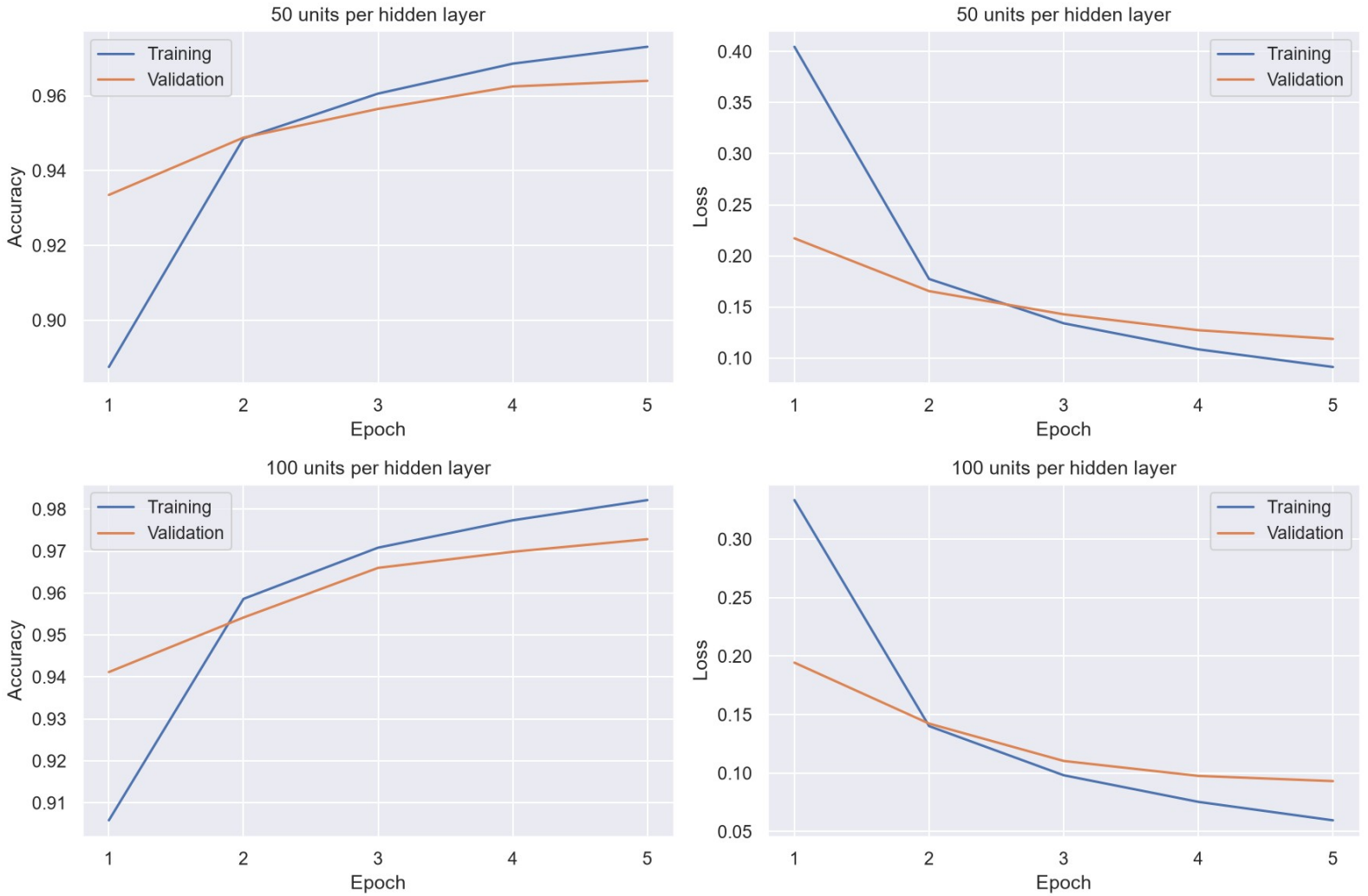
Hidden units	Parameters	Validation accuracy	Validation loss	Fit time (seconds)
50/50	42,310	96.4000%	0.118696	23.747066
100/100	89,610	97.2833%	0.093159	23.330180

Selected: 100/100

Highest final-epoch validation accuracy. No test-based selection.

Source: results/stage3/neural_network_trials.csv and selected_settings.json. Times include validation during epochs

Learning across five epochs



Accuracy rose.
Loss fell.

Training accuracy exceeded validation accuracy.

Five epochs do not establish that overfitting is absent.

Unchanged Stage 3 figure. Each row is one network. Left: accuracy. Right: sparse categorical cross-entropy loss

Evaluation methods

Measure	Meaning
Accuracy	Correct predictions / all predictions
Precision	TP / (TP + FP)
Recall	TP / (TP + FN)
F1-score	2 × precision × recall / (precision + recall)
Macro average	Equal-weight mean across the ten digits

10,000 official test images. Frozen models. Unchanged preprocessing.

TP, FP and FN refer to one digit versus all others. Confusion matrices use actual rows and predicted columns

Final test comparison

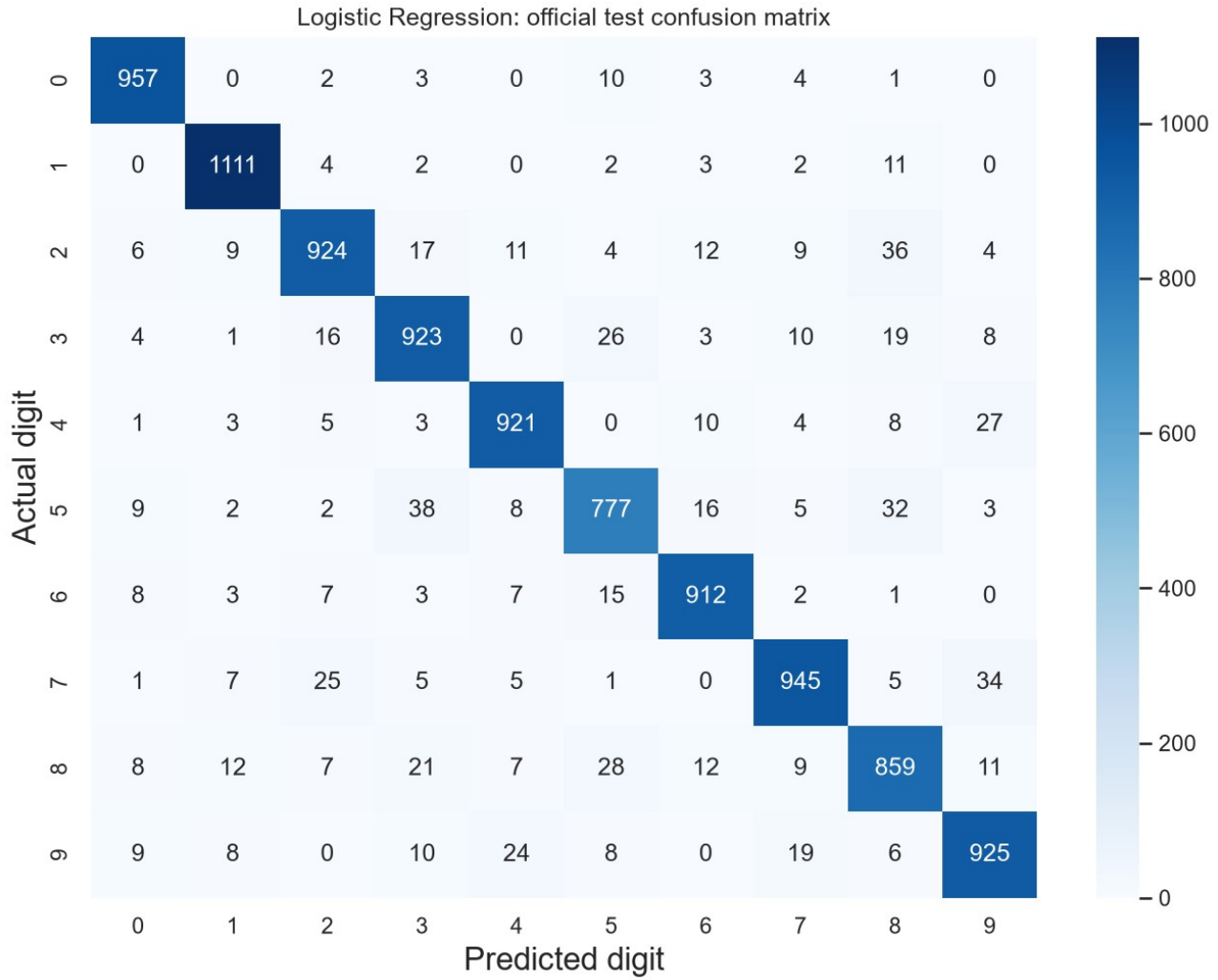
Measure	Logistic Regression	Neural Network
Test accuracy	92.54%	97.53%
Macro precision	0.924432	0.975181
Macro recall	0.924348	0.974954
Macro F1	0.924303	0.975045
Correct predictions	9,254	9,753
Incorrect predictions	746	247
Recorded fit (s)	52.845251	23.330180

+4.99 percentage points. 499 fewer errors.

Timing scopes differ. Both network candidates totalled 47.077246 s.

Source: Stage 4 final comparison. NN fits include validation each epoch. LR fit excludes later validation prediction

Logistic Regression confusion matrix



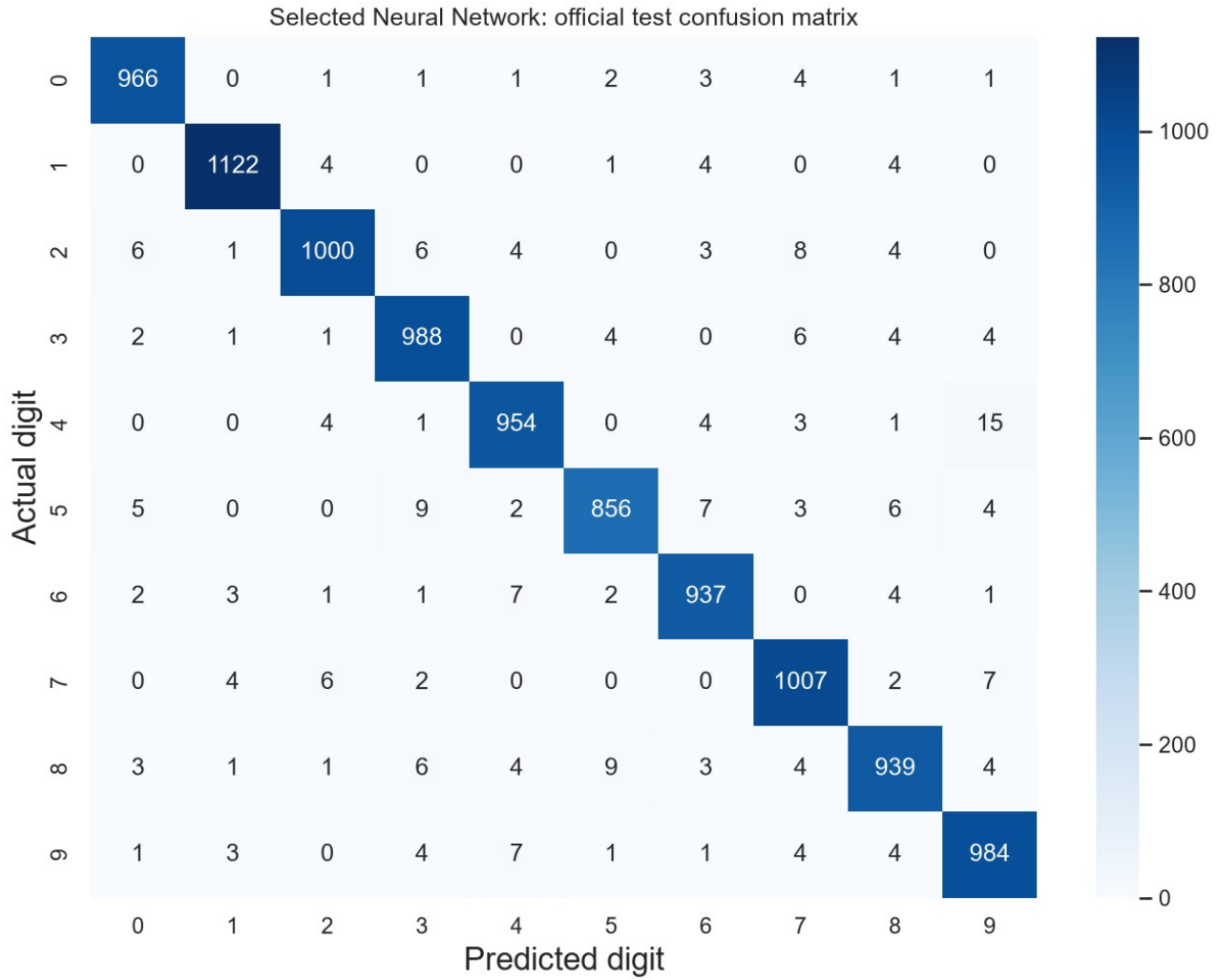
9,254 correct
746 incorrect

Frequent mistakes

5 predicted as 3: 38
2 predicted as 8: 36
7 predicted as 9: 34

Actual digits form rows. Predictions form columns. Matrix total: 10,000. Source: unchanged Stage 4 figure

Neural Network confusion matrix



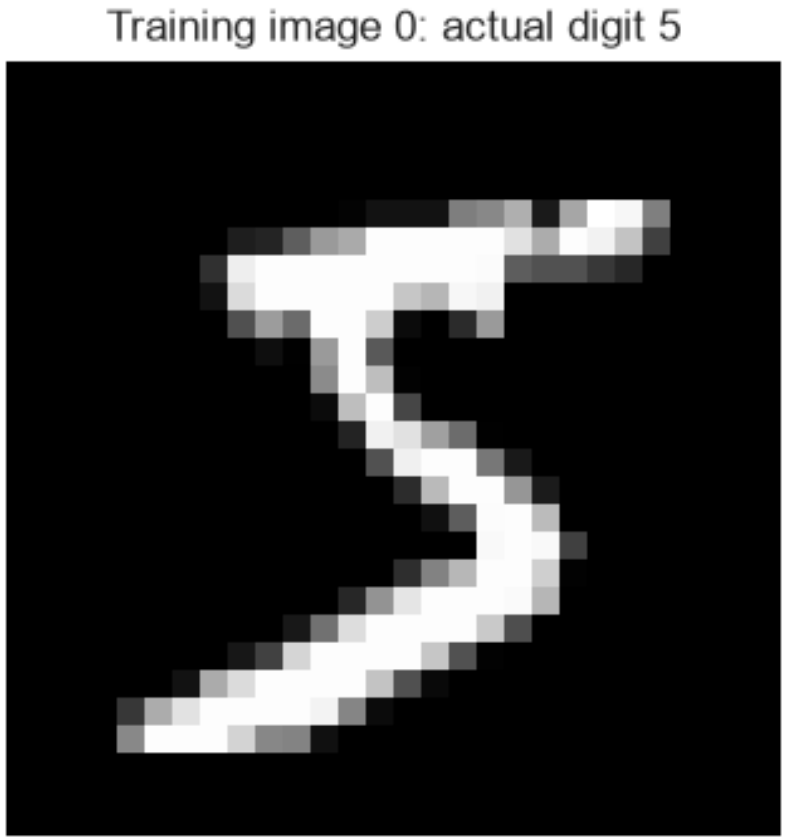
9,753 correct
247 incorrect

Frequent mistakes

4 predicted as 9: 15
5 predicted as 3: 9
8 predicted as 5: 9

Same 10,000 test examples. Counts are directional. Similar strokes are a possible explanation, not a proven cause

Handwritten digit demonstration



MNIST_Demonstration.ipynb
sample_index = 0

Actual digit: 5
Logistic Regression: 5
Neural Network: 5

Changing the index selects another image.
Both models receive the same pixels.

Real Stage 6A example. Original training images illustrate prediction. They do not provide a new generalisation test

Limitations, recommendations and conclusion

The selected network achieved 97.53% versus 92.54% on this MNIST test set.

Limitations

One split and one run per candidate.

Prepared MNIST images only.

Different fit-timing procedures.

Future recommendations

Repeat runs across seeds.

Assess new handwriting with a fresh independent evaluation set.

Measure equivalent timing scopes.

Conclusion applies to this experiment. Full evidence and research references: report/TECHNICAL_REPORT.md